

Getting started



Getting your bearings

A compass will help you get your bearings if you are away from familiar surroundings. Use it in conjunction with the planisphere to locate objects in the night sky.

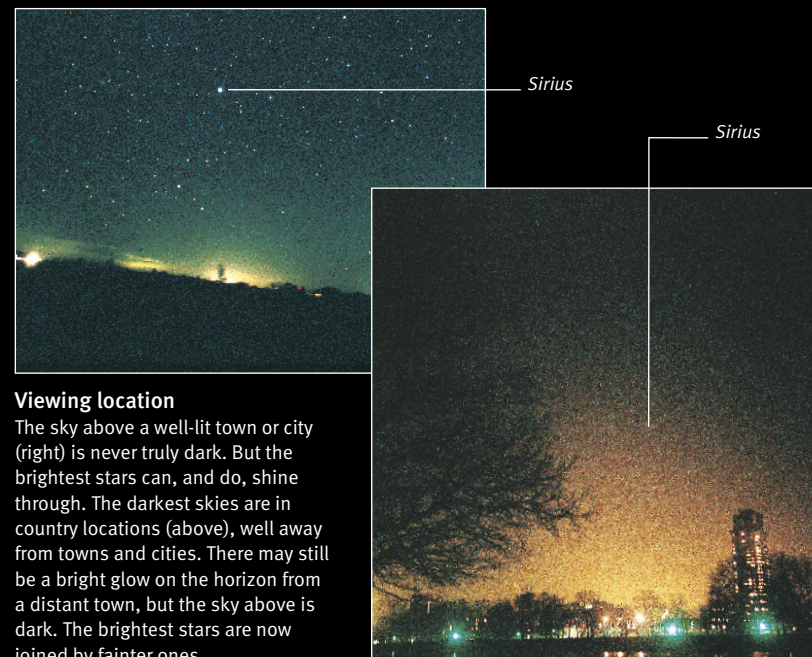
CHECKLIST

- Take warm clothing, even on summer nights.
- Wear a hat in winter; fingerless gloves are useful.
- Carry binoculars around your neck for ease of use.
- Keep your flashlight handy: its soft light keeps your eyes adapted to the dark.
- Use a reclining chair, if possible, for extra comfort.
- Take food and drinks if observing for a long time.
- Make sure your horizon is clear of obstructions.
- Set up your planisphere before going into the dark.

PLANNING AHEAD

The choice of observation site dictates the quality of the sky. Stars, planets, nebulae, and galaxies are best seen in a dark sky away from houses and streetlights. If the sky is clear and moonless, about 300 stars are visible from cities. Only the brightest stand out, which to the absolute beginner can be an advantage: it is the brightest stars that make the constellation patterns. A darker, small-town sky will yield about 1,000 stars; and the darkest rural location, about 3,000. Here the constellation patterns are not so clear, and dim objects are easier to see.

Once you have chosen a location, think about what you need; the checklist (left) will help. Get everything together before going out. A lucky few may not even need to leave the house. At the right location, bright stars, the Moon, and planets can all be observed from a window. Simply turn off the lights and look out and up.



Viewing location

The sky above a well-lit town or city (right) is never truly dark. But the brightest stars can, and do, shine through. The darkest skies are in country locations (above), well away from towns and cities. There may still be a bright glow on the horizon from a distant town, but the sky above is dark. The brightest stars are now joined by fainter ones.

READY TO GO

Any cloud-free night is an observing night, but some are better than others. Use the Monthly Sky Guide (pp.44–69) to find out the phase of the Moon. If the Moon is full, the sky will be flooded with light and stars will be lost from view. Use the Sky Guide and planisphere to plan your viewing.

IDENTIFYING THE STARS

Initially you should aim to identify two or three of the more prominent constellations and the brightest stars. You can build on this base in the nights ahead. Familiarize yourself with the constellation shapes before going outside.

Remember: as Earth spins, our view of a constellation changes as it moves across the sky.

And southern-hemisphere observers should

know that maps of the Moon and constellations are usually oriented for northern-hemisphere observers.

Once outside, give your eyes time to adjust to the dark. Within ten minutes or so, more stars will appear. Before 30 minutes have passed, your eyes will be fully dark-adapted, and a range of stars of different brightness will be discernible. Use the technique of averted vision to observe fainter objects. Look slightly away rather than directly at the object. It will then appear as an image, formed by the sensitive edge of the eye's retina.

Looking skyward

The darkest part of the sky is the highest part, farthest away from the horizon. If possible, lean against a wall when looking up, since this will keep your body steady.



Your view

These two celestial spheres show how Orion looks to observers in the northern and southern hemispheres. Northern observers see Orion with head high, and feet pointing down. By contrast, to the southern observer, Orion appears to be standing on his head.

